

Project Synopsis

QuizPrep — An Interactive Quiz Platform for Interview Preparation

Title

QuizPrep — An Interactive Quiz Platform for Interview Preparation

Overview

QuizPrep is a frontend-only web application designed to help job seekers and students prepare for technical and HR interviews through timed, topic-based quizzes. It includes a personalized dashboard that tracks performance, streaks, and weak areas over time, turning passive question-answering into a structured, data-driven prep routine. All quiz data is served via local/mock JSON, and question generation is optionally augmented using an AI API, making the project easy to run, demo, and extend without a custom backend.

Problem Statement

Most interview prep resources are static PDFs or scattered blog posts with no way to track progress, identify weak topics, or simulate real interview time pressure. QuizPrep solves this by combining structured question banks, timed quiz sessions, AI-assisted question generation, and performance analytics — giving users a clear picture of where they stand and what to study next.

Objectives

- Build a scalable, client-side quiz engine supporting multiple categories (DSA, OOPs, DBMS, OS, CN, System Design, HR/Behavioral)
- Implement a performance dashboard with score history, accuracy, and topic-wise breakdown
- Support timed and untimed quiz modes to simulate real interview/exam pressure
- Integrate an AI API to dynamically generate new questions and explanations
- Demonstrate clean state management, routing, and reusable component architecture
- Persist user progress and preferences entirely on the client (localStorage)

Core Features

- **Quiz Engine** — topic/category selection, difficulty levels (easy/medium/hard), timed and untimed modes
- **Question Bank** — MCQs with instant feedback, explanations, and tagging by topic/company
- **AI-Generated Questions** — on-demand question and explanation generation via an AI API (e.g., OpenAI/Gemini) for topics beyond the static bank
- **Dashboard** — score history, accuracy %, streaks, and topic-wise strength/weakness charts

- **Bookmarks / Review Mode** — save tricky questions to revisit later
- **Authentication (UI only)** — login/signup forms with client-side validation and protected dashboard routes
- **Progress Tracking** — daily streaks, attempted vs. total questions, revision reminders
- **Responsive Design** — mobile-first layout, usable during commute/on-the-go prep
- **Dark/Light Theme** — user-toggable theme persisted across sessions

Tech Stack

Layer	Technology
Framework	React + TypeScript
Build Tool	Vite
Styling	Tailwind CSS
State Management	Redux Toolkit / Zustand
Routing	React Router
Charts (Dashboard)	Recharts / Chart.js
HTTP Client	Axios / Fetch API
Data Source	Local / Mock JSON (static question bank)
AI Integration	OpenAI API / Gemini API (client-side calls for question generation)
Persistence	Browser localStorage (progress, theme, bookmarks)
Testing	Vitest + React Testing Library
Deployment	Vercel / Netlify

Scope & Limitations

- No real backend, database, or authentication server — all user data lives client-side
- Question bank size limited to what is manually curated/seeded plus AI-generated additions at runtime
- AI-generated content depends on third-party API availability and may require an API key
- Leaderboard and multi-device sync are out of scope (would require a backend)
- Intended as a portfolio/demo project, not a production-grade exam platform

Target Outcomes

- A polished, deployable demo suitable for a portfolio and LinkedIn showcase

- A reusable quiz-engine component library (question cards, timers, progress charts, modals)
- Demonstrated ability to design dashboards, manage complex state, and integrate third-party AI APIs
- Clean Git history with meaningful commits, suitable for resume/interview discussion